8-BIT
ENGINEERING

# 8-BIT ENGINEERING

COMPANY PORTFOLIO | FIRMENPORTFOLIO

Your B2B Engineering Partner from Germany
17 Years of Custom Electronics Experience

*Ihr B2B-Engineering-Partner aus Deutschland*
*17 Jahre Erfahrung in kundenspezifischer Elektronik*

PCB Design | Firmware | IoT Systems | 3D Modeling | Prototyping

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EST. 2009

# About Us */ Über Uns*

**8-Bit Lab Engineering** is a specialized B2B engineering company based in Bad Teinach, Baden-Württemberg, Germany. With **17+ years of hands-on experience** since 2009, we deliver end-to-end custom electronics solutions — from initial circuit design and PCB layout to production-ready firmware. We work as a flexible, project-based engineering partner: no overhead, no long-term contracts, just reliable delivery. Our core expertise spans MCU programming, communication protocols (UART, I2C, SPI), cellular IoT (nRF9161), mesh networking, and motor control systems.

***8-Bit Lab Engineering*** *ist ein spezialisiertes B2B-Ingenieurbüro mit Sitz in Bad Teinach, Baden-Württemberg. Mit über* ***17 Jahren praktischer Erfahrung*** *seit 2009 liefern wir durchgängige, kundenspezifische Elektroniklösungen — von der Schaltplanung und dem PCB-Layout bis zur produktionsreifen Firmware. Flexibel, zuverlässig und EU-ansässig.*

| 17+ | 40+ | 100% | B2B |
|---|---|---|---|
| Years Experience | Projects Delivered | Success Rate | Focused Partner |
| *Jahre Erfahrung* | *Projekte geliefert* | *Erfolgsquote* | *Fokussierter Partner* |

## Why Choose Us */ Warum wir*

### Rapid Prototyping
From concept to working prototype in weeks, not months.
*Schnelle Prototypen: Vom Konzept zum funktionierenden Prototyp in Woche*

### Full-Stack Hardware
Circuit design, PCB layout, firmware — all from one source.
*Full-Stack Hardware: Schaltungsdesign, PCB-Layout, Firmware — alle*

### Startup Friendly
We understand lean budgets. Flexible project-based pricing.
*Startup-freundlich: Wir verstehen knappe Budgets. Flexible projektbasierte P*

### EU-Based & Reliable
Based in Germany. Fast communication, no timezone issues.
*EU-ansässig & zuverlässig: Sitz in Deutschland. Schnelle Kommunika*

# Our Services

*/ Unsere Dienstleistungen*

## PCB Design & Layout

*PCB-Design & Layout*

Multi-layer PCB design from schematic capture to Gerber output. Optimized for manufacturability (DFM), signal integrity, and compact form factors.

Altium | KiCad | 2-8 Layer | DFM

## Firmware Development

*Firmwareentwicklung*

Bare-metal and RTOS-based firmware for ARM Cortex-M, AVR, ESP32, and Nordic platforms. Communication protocols, sensor integration, and OTA updates.

C/C++ | nRF SDK | Arduino | FreeRTOS

## IoT & Connectivity

*IoT & Konnektivität*

End-to-end IoT solutions: cellular (LTE-M/NB-IoT via nRF9161), mesh networking (ESP-NOW, Zigbee), BLE, and cloud integration.

nRF9161 | ESP-NOW | BLE | MQTT

## Circuit & Schematic Design

*Schaltungsdesign*

From power supply design to analog signal conditioning and motor drive circuits. Component selection, BOM optimization, and design review.

BLDC Control | Analog | Power | EMC

## 3D Modeling & Enclosures

*3D-Modellierung & Gehäuse*

Custom enclosure design, mechanical integration, and 3D-printed prototypes for your hardware projects.

Fusion 360 | SolidWorks | 3D Print

## Software Development

*Softwareentwicklung*

Desktop and embedded software in C#, Python, and C/C++. Configuration tools, test utilities, and device communication interfaces.

C# | Python | .NET | Serial/USB

# Technology Stack / *Technologie-Stack*

## Microcontrollers / MCUs

- nRF9161 (Nordic)
- nRF52840
- ESP32 / ESP32-S3
- STM32 (ARM Cortex-M)
- ATmega / AVR
- Arduino Ecosystem
- SAMD21/51
- RP2040

## Communication Protocols

- UART / Serial
- I2C / SPI
- ESP-NOW (Mesh)
- BLE (Bluetooth LE)
- LTE-M / NB-IoT
- MQTT / HTTP
- CAN Bus
- Modbus

## Software & Tools

- C / C++ (Embedded)
- C# / .NET
- Python
- nRF Connect SDK
- Arduino IDE
- PlatformIO
- VS Code
- Git

## Design & CAD

- KiCad / Altium
- Fusion 360
- SolidWorks
- PCB DFM Review
- 3D Printing (FDM/SLA)
- Gerber / BOM
- SPICE Simulation
- ERC/DRC

**Need a specific technology?**

We continuously expand our stack. Contact us to discuss your project requirements.

*Wir erweitern kontinuierlich unseren Stack. Kontaktieren Sie uns, um Ihre Anforderungen zu besprechen.*

# Selected Projects / *Ausgewählte Projekte*

## 01 GPS Animal Tracker

*GPS-Tier-Ortungsgerät*

IoT | Cellular | Embedded

A compact GPS tracking device designed for livestock and animal monitoring. Leveraging cellular connectivity (LTE-M via nRF9161) for real-time position reporting in rural environments with minimal infrastructure. Low-power design for extended battery life in outdoor conditions.

**KEY FEATURES:**

- >> Real-time GPS tracking via cellular network
- >> Ultra-low power consumption for months of battery life
- >> Rugged design for outdoor/farm environments
- >> Cloud dashboard for position monitoring

nRF9161 | GPS/GNSS | LTE-M | Low Power | C Firmware | Custom PCB

## 02 Brushless Motor Controller

*Bürstenloser Motorregler*

Motor Control | Power Electronics

Custom BLDC motor controller with precise speed and torque management. Features advanced commutation algorithms, current sensing, and thermal protection. Designed for industrial reliability with configurable parameters via serial interface.

**KEY FEATURES:**

- >> FOC / 6-step commutation algorithms
- >> Overcurrent and thermal protection
- >> Serial configuration interface
- >> Compact, efficient power stage design

ARM Cortex-M | BLDC Control | PWM | Curre | C/C++ | Custom PCB

## 03 ESP-NOW Wireless Dongle

*ESP-NOW Funk-Dongle*

Wireless | IoT | Child Safety

A toddler safety connection system using ESP-NOW protocol for ultra-fast, low-latency wireless communication. Parent unit and child unit pair automatically with proximity alerts and range monitoring. Zero-config mesh communication without WiFi infrastructure.

**KEY FEATURES:**

- >> Ultra-low latency peer-to-peer wireless
- >> Automatic pairing and range monitoring
- >> Battery-optimized for all-day use
- >> Child-safe enclosure design

ESP32 | ESP-NOW | BLE | Battery Mgmt | Arduino | 3D Enclosure

# Selected Projects / *Fortsetzung*

## 04 Agricultural Mesh GPS System

Agriculture | Mesh | Autonomous

*Landwirtschaftliches Mesh-GPS-System*

A mesh-networked GPS positioning system for autonomous farming equipment. Multiple nodes communicate via ESP-NOW mesh to provide precise field coverage mapping and coordinate autonomous vehicle movements. Designed for large-scale agricultural operations where cellular coverage is limited.

**KEY FEATURES:**

- >> Self-healing mesh network across fields
- >> RTK-capable GPS for cm-level precision
- >> Solar-powered autonomous nodes
- >> Cloud integration for fleet management

ESP-NOW Mesh | GPS/RTK | nRF9161 | Solar Pc

## 05 Smart Desk Controller

Industrial | Display | B2B Partner

*Intelligenter Tischregler*

A height-adjustable desk controller developed as a B2B project for a partner company. Features a 2-inch OLED display for real-time height readout, memory positions, and smooth motor control. Intuitive up/down interface with programmable presets and safety features.

**KEY FEATURES:**

- >> 2" OLED display with real-time feedback
- >> Programmable height memory presets
- >> Smooth acceleration/deceleration curves
- >> Anti-collision safety system

STM32 / AVR | 2" OLED (I2C) | Motor Driver

## Have a similar project in mind?

We'd love to hear about it. Let's discuss your requirements.

*Ein ähnliches Projekt? Lassen Sie uns Ihre Anforderungen besprechen.*

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# How We Work */ Unser Arbeitsprozess*

### 1 Discovery & Requirements
*Bedarfsanalyse*

We start with a free consultation to understand your project goals, technical requirements, and timeline. No jargon — just clear communication.

### 2 Concept & Schematic Design
*Konzept & Schaltplanentwurf*

We create the circuit schematic, select components, and present a clear design concept. You review and approve before we move forward.

### 3 PCB Layout & Prototyping
*PCB-Layout & Prototyping*

Professional PCB layout with DFM review. We order prototypes, assemble, and verify functionality with rigorous testing.

### 4 Firmware Development
*Firmware-Entwicklung*

Custom firmware tailored to your hardware. From bootloader to application layer, including communication stacks and sensor drivers.

### 5 Testing & Validation
*Test & Validierung*

Comprehensive testing: functional, environmental, and edge-case scenarios. We deliver documented test results and verified hardware.

### 6 Delivery & Support
*Lieferung & Support*

Complete handover: design files, source code, BOM, and documentation. Ongoing support available for production scaling.

## Typical Timeline: 4-12 Weeks

From first call to working prototype — depending on complexity.

*Vom ersten Gespräch zum funktionierenden Prototyp — je nach Komplexität.*

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# Let's Build Something Great

*Lassen Sie uns etwas Großartiges bauen*


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**PCB Design · Embedded Firmware · IoT Systems · Prototyping**

Project-based B2B collaboration — no employee, no overhead.

*Projektbasierte B2B-Zusammenarbeit — kein Festangestellter, kein Overhead*

Scan to visit www.8bitlab.de

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